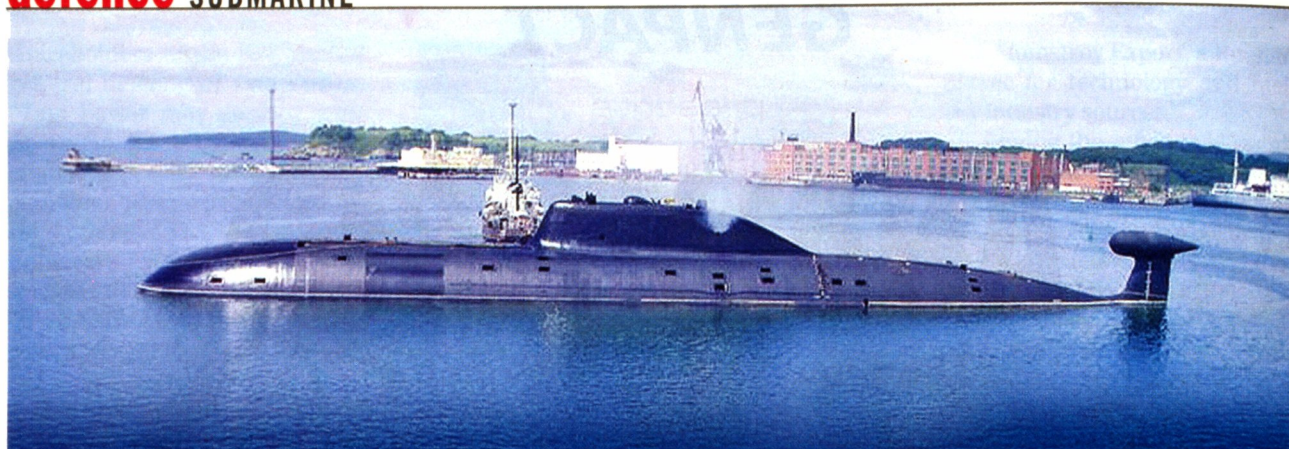




The Sub Total

The navy sets Republic Day and Independence Day dates for launching and commissioning two nuclear submarines

India's strategic establishment has begun the countdown to two important milestones. In late June, a modified Akula-II class nuclear-powered attack submarine, the *Chakra*, began harbour acceptance trials at the Komsomolsk-on-Amur shipyard in the Russian far east. In the next few months, the Russian submarine, formerly the *Nerpa* (Seal), is to sail into the Pacific Ocean off Vladivostok for full-fledged sea trials with a Russian crew. If the trials are successfully concluded, say officials, the *Chakra* will be commissioned at Vladivostok with an Indian crew who will sail it to India by August 15, 2009.

The choice of date is not accidental. The second milestone, too, has been timed with a national event. On January 26, 2009, the sluice gates of an enclosed drydock in Visakhapatnam are to be opened and the world will take its first look at India's first nuclear-powered submarine, the Advanced Technology Vessel (ATV), as it enters the waters. Nuclear submarines are powered by a nuclear reactor which produces enormous heat that runs steam turbines. They can stay underwater almost indefinitely and stealthily launch ballistic missiles from under the sea.

The national maritime doctrine unveiled last year calls a nuclear-armed missile submarine the "most credible

of all arsenals in a second strike" and "most preferred" for small nuclear forces". The aim is to field three submarines equipped with nuclear-tipped ballistic missiles—the so-called third leg of the nuclear triad comprising air, land and sea-launched weapons—on "deterrent patrols" (to deter a potential adversary from launching a nuclear first strike) by 2015.

India's ATV is based on the Russian Akula-I class submarine, but is powered by a single indigenously-built nuclear reactor and equipped with 12 K-15 ballistic missiles or 16 of the Indo-Russian BrahMos supersonic cruise missiles. Three ATVs are under construction, and hull fabrication and integration of the nuclear reactor on the first submarine have been completed. Like the *Chakra*, it will begin harbour trials in a specially-made, enclosed anchorage in the Ship Building Centre at Visakhapatnam. Here, all the on-board systems, including the reactor and steam turbines, are to be fully tested

FIRST LOOK: The *Chakra* on harbour trials at a Russian shipyard

before it can sail out for sea trials in the Bay of Bengal. ATV is expected to join the navy after a two-year shake-down period.

Due to its design commonality with ATV, the *Chakra* is being leased primarily to train crews to man ATVs. With its cruise missiles and torpedoes, it will also be used for sea-denial missions in the Indian Ocean. The under-construction 12,000-tonne *Chakra* was to be completed and leased to India for 10 years under a \$650-million deal signed in

2004. It was to be inducted into the navy on August 15 this year, but was delayed by a year, following technical snags.

"Project India", as the classified lease programme is called, seems to be back on track after being seemingly caught in last year's freeze in Indo-Russian defence ties over escalating costs of the aircraft carrier *Vikramaditya's* refit.

by Sandeep Unnithan

DEBUT PLANS

- ▶ **ATV to be launched in Visakhapatnam on Jan 26, 2009.**
- ▶ **The *Chakra* to be inducted into the navy on Aug 15, 2009.**
- ▶ **ATV is a modified Akula-I class submarine.**
- ▶ **It will be the first submarine equipped with BrahMos missiles and the indigenous 700 km-range K-15 missiles.**
- ▶ **The *Chakra* is a modified Akula-II class submarine.**